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Northeast Fisheries
Science Center



Atlantic Sea Scallop Research Track Peer Review Meeting

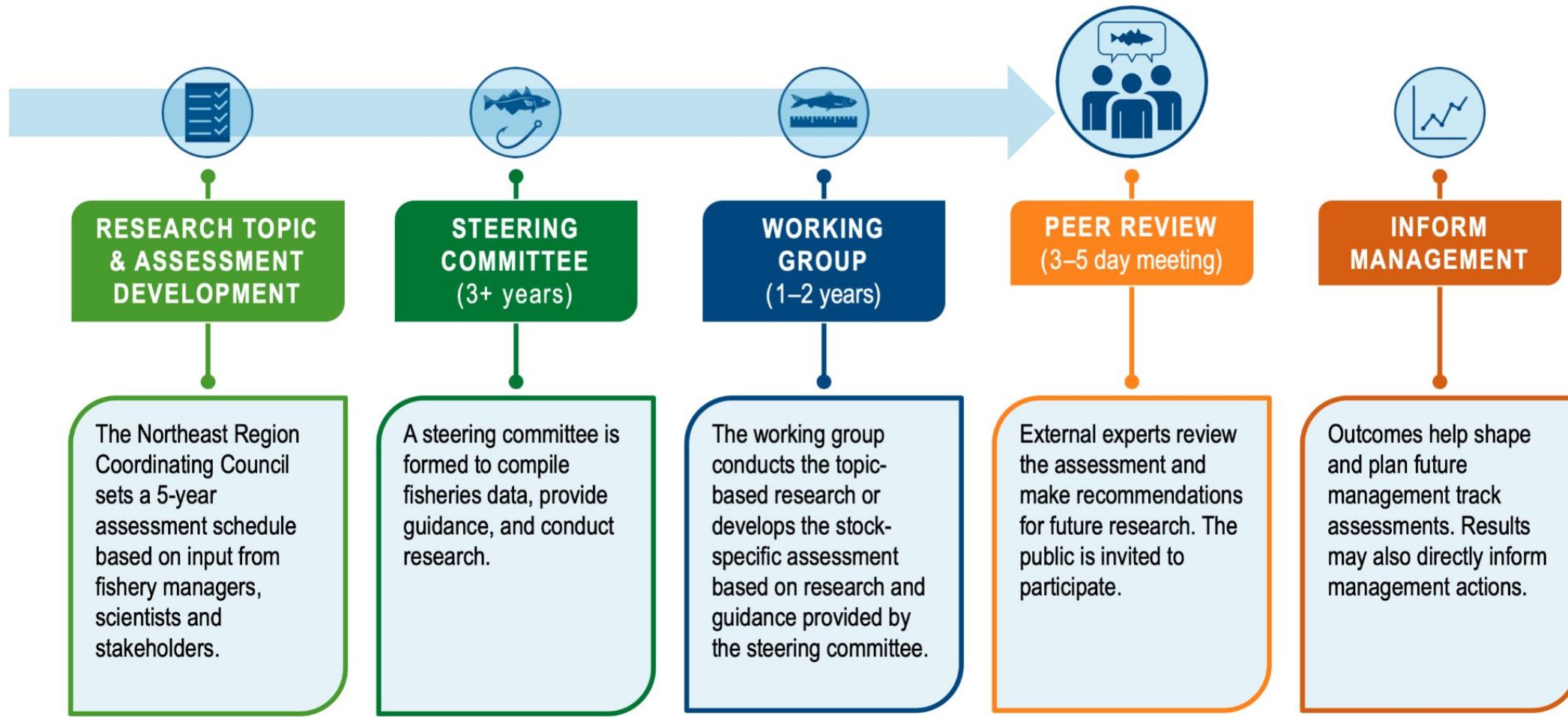
April 21-24, 2025

NRCC Assessment Meeting Code of Conduct

- NRCC committed to providing all participants a safe, productive, inclusive, and welcoming environment
- Code applies to all meeting participants, in person and virtual
- All participants are expected to maintain a professional and cordial work environment
- More details can be found in the [Code of Conduct](#) on the [NRCC webpage](#)



NRCC Stock Assessment Process



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Review Panel Responsibilities

Objective: Determine whether the completed RT assessment is technically sufficient to:

1. Evaluate stock status
2. Evaluate new data streams and/or model changes
3. Successfully address the assessment TORs

Panelists should:

- Focus discussions on work that is within the scope of the RT (i.e., under one of the eight TORs)
- Address whether each TOR was successfully completed
- Provide prioritized recommendations for future improvements



Atlantic Sea Scallop Research Track WG

Members

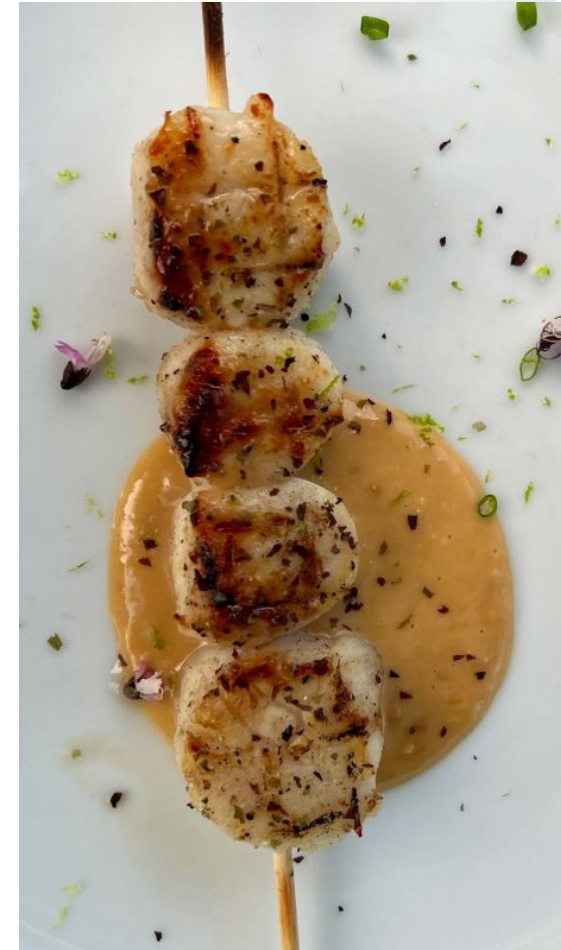
- Patrick Sullivan, Chair – Cornell University
- Dvora Hart, Assessment Lead – NEFSC
- Jui-Han Chang, Co-Assessment Lead – NEFSC
- Jessica Blaylock – NEFSC
- Adam Delargy – UMass Dartmouth
- David Keith – Canada DFO
- Amber Lisi – UMass Dartmouth
- Jonathon Peros, NEFMC
- David Rudders – VIMS

Chair Invited Participants

- Joseph Caracappa – NEFSC
- Kim Hyde – NEFSC
- Carl Huntsburger - ME DMR

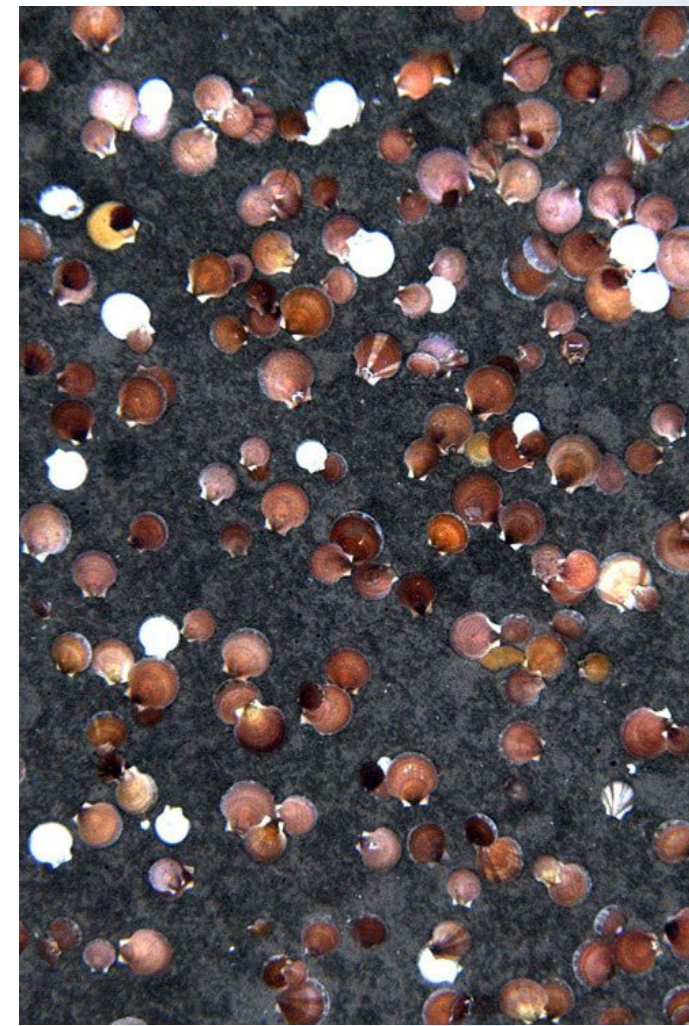
Acknowledgements

- Data providers
- Scallop constituents



Atlantic Sea Scallop RT Schedule

- **Research Track Working Group:** January 2023 through April 2025
- **Research Track Assessment Peer Review:** April 21-24, 2025
- **Future:**
 - Implement new methods via annual scallop management support
 - Continue to develop science advancements for scallops based on priority research recommendations via multiple pathways



Peer Review Panel

- Dr. Thomas Miller, Chair
- Dr. Martin Cryer, CIE
- Dr. Yong Chen, CIE
- Dr. Noel Cadigan, CIE



Key NEFSC Support

- Kristan Blackhart, Population Dynamics Branch Chief
- Brian Hooper, Assessment Process Lead



Research Track Terms of Reference

1. Identify relevant ecosystem and climate influences on the stock
2. Estimate catch from all sources
3. Present the survey data used in the assessment
4. Use the appropriate assessment approach to estimate F , R , and B
5. Update or redefine Biological Reference Points and provide updated Stock Status determinations
6. Define appropriate methods for producing projections
7. Review, evaluate, and report on the status of research recommendations
8. Develop a backup assessment approach



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Review Agenda

<i>Day</i>	Monday April 21	Tuesday April 22	Wednesday April 23	Thursday April 24
<i>Mode</i>	Hybrid NEFSC Clark Conference Room	Hybrid NEFSC Clark Conference Room	Hybrid NEFSC Clark Conference Room	Hybrid NEFSC Clark Conference Room
<i>Time</i>	9:00 am – 5:00 pm	9:00 am – 5:00 pm	9:00 am – 5:00 pm	9:00 am – 1:00 pm
<i>Topics</i>	Background TORs 1 & 2	TORs 3 & 4	TORs 5, 6, 8	TOR 7

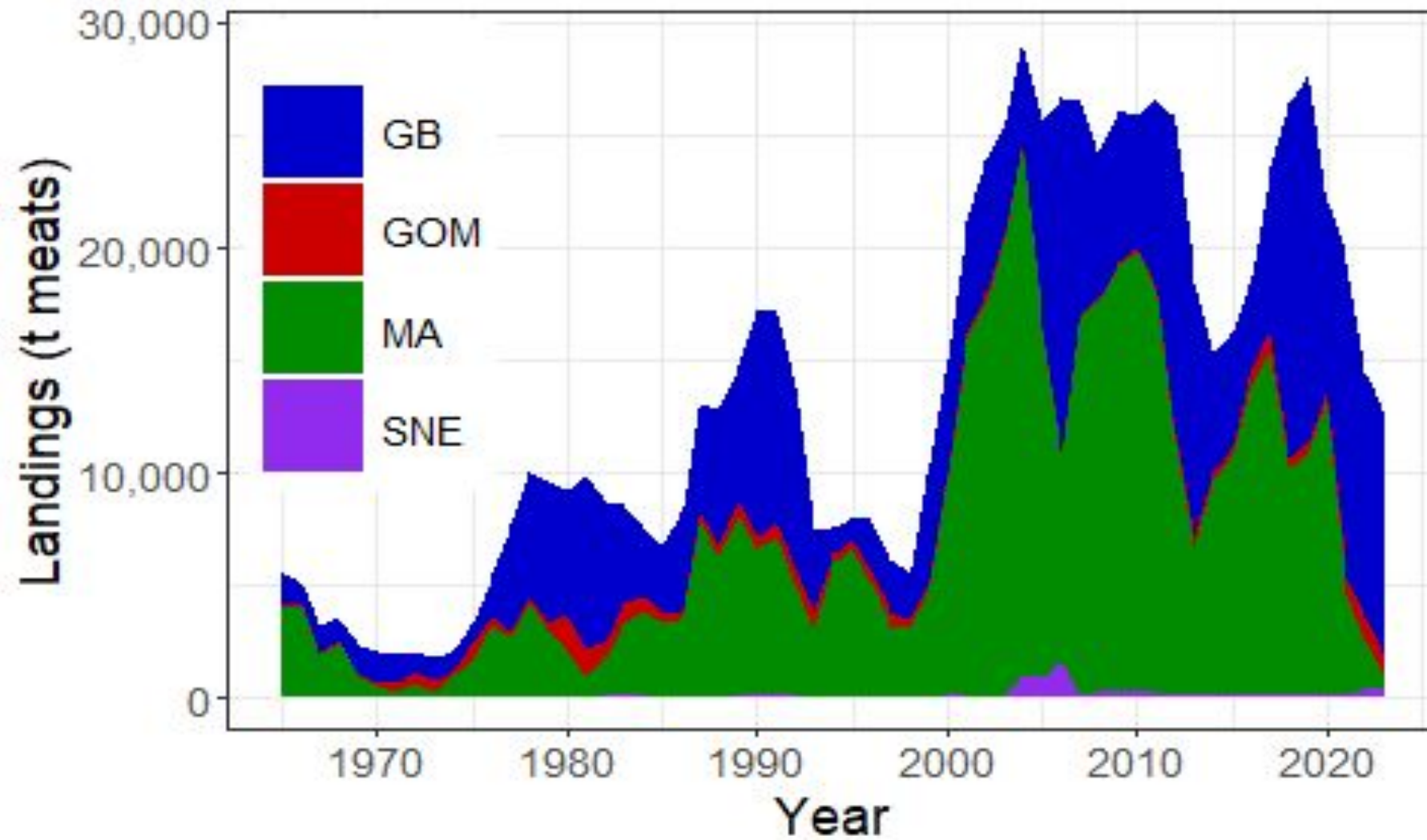
TOR 1: Ecosystem Influences

- Temperature influences expanding northward and to inshore areas throughout the Mid-Atlantic
- Growth/weight at shell height tends to increase at shallower depths
- The sea star *Astropecten americanus* appears to be reducing or excluding scallops from deeper areas of the Mid-Atlantic and is expanding its range
- Scallop parasites decrease meat weights, while their impacts on scallop population abundance remain poorly understood
- Shell blister disease in scallops may be associated with warming temperatures and are correlated with higher mortality rates



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TOR 2: Catch Data



TOR 3: Survey Data

- Scallop survey data collected collaboratively using three methods by multiple regional partners
- Dredge surveys: NEFSC (1975-present), VIMS (1999-present)
- Drop camera survey: SMAST (2003-present)
- Habcam towed camera surveys (2012 - present)



TOR 4: Assessment Approach

- Scallops use bespoke size-structured models (Catch-at-Size-Analysis, or CASA) to assess three regional sub-populations from 1975-2023:
 - Mid-Atlantic
 - Georges Bank Open
 - Georges Bank Closed
- Advancement: Growth transition matrices estimated from growth increment data outside of CASA
- Advancement: Natural mortality estimated within CASA model by year and size for juveniles and/or adults or annually without varying by size

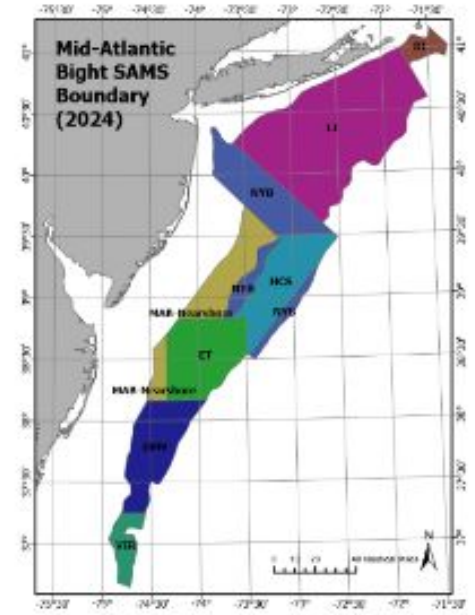
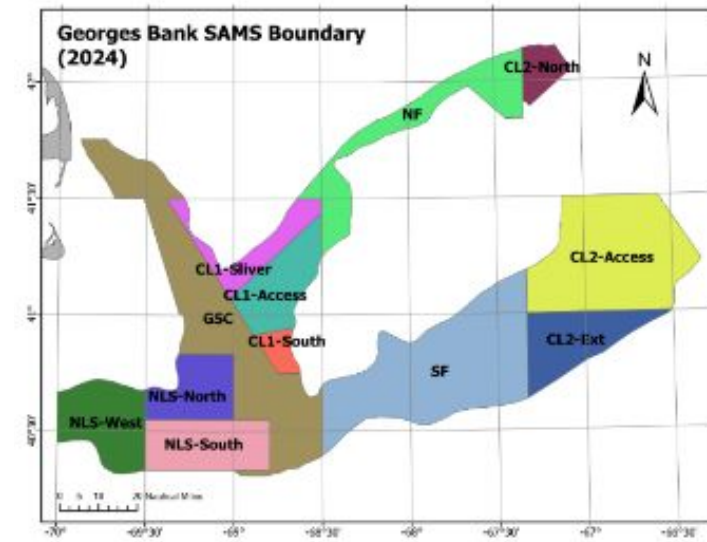


TOR 5: Biological Reference Points

- Reference points calculated using the Stochastic Yield Model (SYM) separately for the Mid-Atlantic and Georges Bank, then combined to a single value for the entire stock
- $F_{MSY} = 0.49$
- $B_{MSY} = 93,282$ mt meats
- $B_{2023}/B_{MSY} = 75\%$; stock is not overfished
- $F_{2023}/F_{MSY} = 67\%$; stock is not experiencing overfishing

TOR 6: Projections

- Projections developed using the Scallop Area Management Simulator (SAMS), a spatial forecasting model that divides the scallop resource into a number of subareas
- SAMS models population dynamics similar to CASA and SYM, but for each subarea
- SAMS can simulate the rotational management used for the scallop fishery



TOR 7: Research Recommendations

- TOR 7 WP outlines how previous research recommendations have been addressed
- WG provides new research recommendations, including:
 - Fill data gaps by aging older scallop samples
 - Review the performance of area-specific projections
 - Continue to investigate environmental impacts on scallops
 - Continue to analyze the use of AI algorithms to support analysis of images collected using optical survey tools (i.e., Habcam, drop camera)



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TOR 8: Backup Assessment Approach

- **If a CASA model fails** → Use swept area survey estimates of density expanded to suitable habitat areas
- **If a SYM model fails** → Use deterministic calculation of reference points
- **If CASA and SYM models fail** → Revert to a swept area estimate with average catch to determine appropriate exploitation rates

Questions?



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